SOMAVILLE UNIVERSITY

HAYAAT SUPERMARKET MANAGEMENT SYSTEM

MOGADISHU SOMALIA

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A PROPOSAL REPORT SUBMITTED TO *Hayaat supermarket managemnets system* in PARTIAL FULLFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF COMPUTER SCEINCE OF SOMAVILLE UNIVERSITY

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DECLARATION

This thesis is our original work and has not been presented for a degree in any other university.”

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APPROVAL

We certify that this Research report satisfies the partial fulfillment of the requirements for the award of the Bachelor Degree of Computer Science & it at Somaville University in Mogadishu Somalia, and has been carried out by the candidates under our supervision and was submitted with our approval

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DEDICATION

We dedicate this piece of work to our parents they have been source of inspiration, engine of courage and secret of our achievements. We also dedicate it to our sisters and brothers for all the support.

ACKNOWLWDGEMENT

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Contents

CHAPTER ONE

1.0 INTRODUCTION

In today's rapidly evolving retail environment, effective management of supermarket operations is crucial for maintaining competitive advantage and ensuring customer satisfaction. Supermarkets, which are essential components of the retail sector, face increasing demands to streamline their processes, manage inventory, and improve overall service delivery. The complexities of handling a large volume of transactions, stock management, and customer interactions make it imperative for supermarket managers to adopt innovative solutions that can enhance operational efficiency.

This study focuses on the development of a **Supermarket Management System**, aimed at addressing the challenges supermarkets face in terms of inventory control, sales tracking, and customer relationship management. The project seeks to provide a comprehensive software solution that integrates various operational aspects, offering supermarket owners and managers a user-friendly platform for real-time data access and decision-making. By automating key tasks, this system is designed to not only reduce human error but also improve the overall customer experience, thereby contributing to the success and sustainability of supermarket operations.

1.1 BACKGROUND OF THE STUDY

With the rapid development of modern science and technology, computer technology has penetrated into all fields and becomes the necessary tools for various industries, especially the Internet technology promotion and the establishment of the information highway. It makes the IT industry increasingly shows its unique advantages in the market competition (Yongchang and Mengyao, 2015). Into the digital age, there is a huge data information waiting for processing and transmission, which makes the further development and use of the database is particularly urgent. As some small and medium-sized supermarkets in the domestic market, they are falling behind the large and medium-sized supermarkets during the informatization, but for these enterprises' resource management, information storage and processing also shows the urgent need. To adapt to market competition, it requires efficient handling and management methods, so it is indispensable that accelerate the process of the computerization of supermarket (Yongchang and Mengyao, 2015).The essence of a supermarket management system is for an effective automation of the management of a supermarket. (Nnanyerem G. 2018).

The Supermarket Management System is a project that deals with supermarket automation and it includes both purchasing and selling of items. Supermarket management system is the system where all the aspects related to the proper management of supermarket is done. These aspects involve managing information about the various products, staff, managers, customers, billing etc. This system provides an efficient way of managing the supermarket information. Also allows the customer to purchase and pay for the items purchased.This study is based on the sales transaction and billing of items in a supermarket. This study is to produce software which manages the sales activity done in a supermarket, maintaining the stock details, maintaining the records of the sales done for a particular month/year. The users will consume less time in calculation and the sales activity will be completed within a fraction of seconds whereas manual system will make the user to write it down which is a long procedure and so paper work will be reduced and the user can spend more time monitoring the supermarket. The program will be user friendly and easy to use. The system will allow the user when to reorder for a particular goods.The system will display all the items whose name start with the letter selected by the user. He can select out of those displayed. Finally, a separate bill will be generated for each customer. This will be saved in the database. Any periodic records can be viewed at any time. If the stock is not available, the supermarket orders and buys from a prescribed vendor. The amount will be paid by deducting the total amount acquired in the sales activity. Admin provides a unique username and password for each employee through which they can login to proceed on their daily activities.

1.2 STATEMENT OF THE PROBLEM

Manual management of the supermarket encounter countless problem which varies from time consumption, poor communication, problem of physical count, daily purchase problem and ordering of supplies. Manual management of a supermarket slow down the updating of daily activities, also it can cause a misunderstanding in a supermarket operation process i.e., poor communication. The use of manual management system will always require either the presence either the supermarket owner or the employee to carry out stock taking which is always time consuming, operating with manual management system may cause an injury for the supermarket if transaction record is lost or misinterpret.

1.3 RESEARCH QUESTION

Here are clear research questions for a Supermarket Management System thesis:

1. How can a supermarket management system improve the efficiency of inventory tracking and management?
2. What are the key functionalities required in a supermarket management system to streamline sales and billing processes?
3. How can customer data management in a supermarket system enhance customer satisfaction and loyalty?
4. What role does a supermarket management system play in optimizing supplier and stock management

1.4 PURPOSE OF THE PROJECT

The purpose of this project is to design and develop an efficient Supermarket Management System that streamlines and automates the day-to-day operations of a supermarket. In today's competitive retail environment, supermarkets face challenges in managing inventory, processing sales transactions, tracking customer data, and maintaining accurate financial records. Manual systems are often time-consuming, prone to human error, and inefficient in handling large volumes of data.

This system aims to address these challenges by providing a comprehensive software solution that facilitates effective inventory management, sales processing, supplier management, customer management, and report generation. The system will ensure real-time updates of stock levels, accurate billing processes, and easy access to sales and financial reports. It will also improve customer service by enabling faster checkout processes and better data management.

By implementing this supermarket management system, supermarket owners and managers can reduce operational costs, minimize stock wastage, improve decision-making through detailed reporting, and enhance overall productivity. Ultimately, the system will help modernize supermarket operations, making them more reliable, scalable, and responsive to both business needs and customer expectations.

1.5 PROJECT OBJECTIVES

- **i.** To know when to order for more goods, keep status and updates of transactions, thereby helping progress level,
- **ii.** To determine stock taking and managerial decisions.
- **iii.** To help the Administrator control different branches from one place.
- iv. To come from old vission system like using books and using new system like supermarket management system

1.6 SCOPE OF THE STUDY

The scope of the project is to develop online Supermarket management system which will service the client.

There are three types of dimension scope which mentioned below

1. Time scope of project should be Aprill 30th up to May 20-30th 2024

2. Geography scope will be Hodan district, is one main district at Banadir region

3. Content of scope. The content scope will be Supermarket management system

is to design and develop by using PHP as front end and MySQL (xampp) as back end

1.7 SAGNIFANCE IF THE STUDY

The online supermarket management system is significant to humanity in the following ways:

i. The management system will be of great benefit to supermarkets and it will improve the managerial and administrative strength of the business and move the business forward to meet the demand of times and globalization in this era of technology.
ii. To explore the challenge being faced by the manual system.
iii. For easy record of goods and proper identification.
iv. This piece of work will add to the already existing literature review and act as a reference work for future researchers

1.8 PROJECT ORGANIZATION

Here is the Systematic writing of our final thesis chapters guided us to follow the regulation of each chapter outlined its description of methodology writing as the following steps:

17.1. Chapter I- Introduction

This chapter will present briefly main concepts of background about the problem of Online Supermarket Management system including with it's of objectives, scope, and significance of this project, expected outcome and how the project will be organized.

1.7.2. Chapter II- Literature Review

This chapter will present an overview of the project in details and discusses the relevant

literature review of Supermarket managemnt system .and the gap analysis between

existing system and proposal system for using tabulated table.

1.7.3. Chapter III- Requirement analysis-This chapter obviously will focus on requirement analysis system for utilizing process modeling including data flow diagram, (DFD) and unified modeling language (UML) and as well as requirement specification system which will be a part of this research like data gathering and feasibility study which to have develop proposed system of Supermarket Management System

CHAPTER TWO

LITERATURE REVIEW

2.0 INTRODUCTION

A supermarket is a self-service shop offering a wide variety of food, beverages and household products, organized into sections. This kind of store is larger and has a wider selection than earlier grocery stores, but is smaller and more limited in the range of merchandise than a hypermarket or big-box market (Wikipedia). In everyday U.S. usage, however, "grocery store" is synonymous with supermarket (oxford dictionary) and is not used to refer to other types of stores that sell groceries.

The supermarket typically has places for fresh meat, fresh produce, dairy, deli items, baked goods, etc. Shelf space is also reserved for canned and packaged goods and for various non food items such as kitchenware, household cleaners, pharmacy products and pet supplies. Some supermarkets also sell other household products that are consumed regularly, such as alcohol (where permitted), medicine, and clothing, and some sell a much wider range of nonfood products: DVDs, sporting equipment, board games, and seasonal items.

According (Meyer Z. 2017). A larger full-service supermarket combined with a department store is sometimes known as a hypermarket. Other services may include those of banks, cafés, childcare centers/creches, insurance (and other financial services), mobile phone services, photo processing, video rentals, pharmacies, and gas stations. If the eatery in a supermarket is substantial enough, the facility may be called a "grocerant", a blend of "grocery" and "restaurant"

The traditional supermarket occupies a large amount of floor space, usually on a single level. It is usually situated near a residential area in order to be convenient to consumers. The basic appeal is the availability of a broad selection of goods under a single roof, at relatively low prices. Other advantages include ease of parking and frequently the convenience of shopping hours that extend into the evening or even 24 hours of the day. Supermarkets usually allocate large budgets to advertising, typically through newspapers. They also present elaborate in-shop displays of products.

2.2 CONCEPTS ABOUT SUPERMARKET MANAGEMENT SYSTEM

Supermarket is a large form of the traditional grocery store, it is a self-service shop offering a wide variety of food and household products, organized into aisles. It is larger in size and has a wider selection than a traditional grocery store, but is smaller and more limited in the range of merchandise than a hypermarket or big-box market (Wikipedia).

According to Ballou, R.H. (1999), Supermarkets proliferated across Canada and the United States with the growth of automobile ownership and suburban development after World War II. Most North American supermarkets are located in suburban strip shopping centers as an anchor store along with other smaller retailers. They are generally regional rather than national in their company branding. Kroger is perhaps the most nationally oriented supermarket chain in the United States but it has preserved most of its regional brands, including Ralphs, City Market, King Soopers, Fry's, Smith's, and QFC.

In Canada, the largest such chain is Loblaw, which operates stores under a variety of regional names, including Fortinos, Zehrs, No Frills, the Real Canadian Superstore, and the largest, Loblaws, (named after the company itself). Sobeys is Canada's second largest supermarket with locations across the country, operating under many banners (Sobeys IGA in Quebec). Québec's first supermarket opened in 1934 in Montréal, under the banner Steinberg's.

According To Yang, B., and Burns, N.D. (2003), said that, in the United Kingdom, self-service shopping took longer to become established. Even in 1947, there were just ten self-service shops in the country. In 1951, ex-US Navy sailor Patrick Galvani, son-in-law of Express Dairies chairman, made a pitch to the board to open a chain of supermarkets across the country. The UK's first supermarket under the new Premier Supermarkets brand opened in Streatham, South London, taking ten times as much per week as the average British general store of the time. Other chains caught on, and after Galvani lost out to Tesco's Jack Cohen in 1960 to buy the 212 Irwin's chain, the sector underwent a large amount of consolidation, resulting in 'the big four' dominant UK retailers of today: Tesco, Asda (owned by Wal-Mart), Sainsbury's and Morrisons.

According to Patton (1990), said that in the 1950s, supermarkets frequently issued trading stamps as incentives to customers. Today, most chains issue store-specific "membership cards," "club cards," or "loyalty cards". These typically enable the card holder to receive special members- only discounts on certain items when the credit card-like device is scanned at check-out. Sales of selected data generated by club cards are becoming a significant revenue stream for some supermarkets.

Pagh, J.D. and Cooper, M.C. (1998), said that it has been determined that the first true supermarket in the United States was opened by a former Kroger employee, Michael J. Cullen, on August 4, 1930, inside a 6,000-square-foot (560 m^2) former garage in Jamaica, Queens in New York City. The store, King Kullen, (inspired by the fictional character

King Kong), operated under the slogan "Pile it high. Sell it low." At the time of Cullen's death in 1936, there were seventeen King Kullen stores in operation. Although Saunders had brought the world self-service, uniform stores and nationwide marketing, Cullen built on this idea by adding separate food departments, selling large volumes of food at discount prices and adding a parking lot.

Rietze S. (2006) debate about the origin of the supermarket, with King Kullen and Ralphs of California having strong claims. Other contenders included Weingarten's Big Food Markets and Henke & Pillot. To end the debate, the Food Marketing Institute in conjunction with the Smithsonian Institution and with funding from H.J. Heinz, researched the issue. It defined the attributes of a supermarket as "self-service, separate product departments, discount pricing, marketing and volume selling".

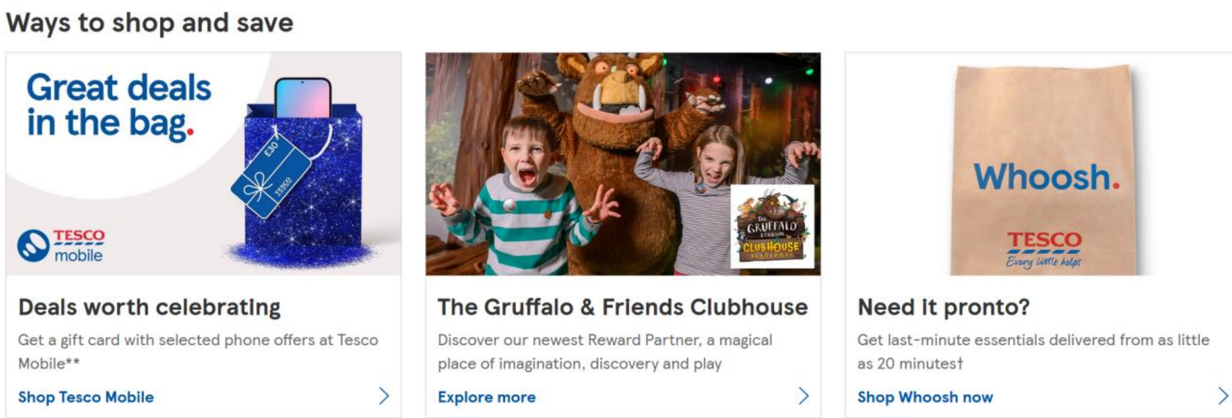
Other established American grocery chains in the 1930s, such as Kroger and Safeway at first resisted Cullen's idea, but eventually were forced to build their own supermarkets as the economy sank into the Great Depression, while consumers were becoming price-sensitive at a level never experienced before. Kroger took the idea one step further and pioneered the first supermarket surrounded on all four sides by a parking lot by Bowersox, C. (2009).

2.3 EXISTTING STSTEMS

The purpose of this research is to find a way implementation that will provide a Solution not only to tourism hospitality as well as resort since it may be study a number of researches in this field by gathering enough information that will enhance the knowledge by comparing relevance research case studies of relating this article entitled ―Online Supermarket management system Management System for web based technology computerized system

2.3.1 Tesco Supermarket

Tesco was founded in 1919 by a man named Jack Cohen in London, UK. It started as a small market stall where Cohen sold surplus groceries. The name "Tesco" was created by combining the initials of his supplier T.E. Stockwell and the first two letters of his own last name – COhen.



Website link: https://www.tesco.com

2.3.2 Hayaat Supermarket

HAYAT MARKET was founded in April 2020 by a group of local investors. headquartered in Mogadishu Somalia.

Hayat Market is the largest selection of quality products You'll get all groceries, household necessities, electronics, toys, and many more

Website link: https://hayatmarket.com

2.3.3 Comparison of Relevance Research among existing systems

No	Feature	Hayat Supermarket	Tesco Supermarket
1	Shopper Marketplace Views	Limited but growing online presence; offline still strong	High online shopper engagement
2	Product Ordering Experience	Online ordering available but slower due to less experience	Fast and smart digital ordering
3	Internet Connectivity	Mixed: Low-speed LAN offline + basic wireless for online use	Wireless WAN internet available everywhere
4	Data Management	Combination: Offline Microsoft Access + online DBMS	Online DBMS (MySQL, SQL Server 2014) / (Oracle 9i)
5	Interface Design	Basic navigation offline; online interface improving	Smooth, user-friendly online interface
6	Login System	Offline login + online login system	Secure online login system
7	User Friendliness	Moderate GUI offline; friendly and improving online	Friendly and interactive online GUI

2.4 GAP ANALYSIS

The system is conducted in order to find out the similarities and the differences among them since the comparison was done based on the following criteria including support modules of table book reservation management System will focus module e-table booking including information security, Connectivity internet technology, promote price satisfaction , DBMS recording , interface design, login system, graphical interface design, mobile application service and user friendliness are all above shown table of comparison among the existing systems which will be similarity to the new proposed system.

2.5 CHAPTER OF SUMMARY

This chapter is about the literature review that discusses the researches of existing systems, which gives a better understanding of furniture store management system. It also shows the similarities and the differences that the existing systems have by comparing them and discusses related things of the current system. It represented the border or limitation of the current system

CHAPTER THREE

REQUIREMENT ANALYSIS

3.0 INTRODUCTION

Requirement analysis is a crucial phase in the development of any information system, including a supermarket management system. It involves identifying, analyzing, and documenting the needs and expectations of the users and stakeholders. The goal is to understand what the users need from the system, both in terms of functionality and performance, and to ensure that these requirements are clearly defined and agreed upon before the design and implementation phases begin.

3.1 USER REQUIREMENT ANALYSIS

User requirements form the foundation of system development. When creating a system from the ground up, it is crucial to define detailed requirements for each process involved. In the case of the **Online Supermarket Management System**, all operational, financial, and legal aspects must be considered, particularly those relating to the retail and supply chain sectors in an e-commerce environment. The internet has revolutionized how businesses operate, turning the world into a global marketplace. With the widespread adoption of the internet, online shopping has become an essential part of modern consumer behavior. Today, users expect convenience, flexibility, and security in their shopping experiences. This trend has led to a growing demand for systems that allow customers to purchase groceries and household items online without visiting physical stores.

An Online Supermarket Management System is designed to address this demand by providing a user-friendly platform where customers can browse products, place orders, and make secure payments—all from the comfort of their homes, at any time of day. The system will simplify the shopping process, offering real-time stock updates, personalized recommendations, and a seamless checkout experience.

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3.1.1 Functional Requirements: of the item.

Keep alert signal for

- List the detail or expired items
- Recording customer information
- Verify the role of the user
- Making product available for customers

3.1.2 Non Functional Requirements:

- System must be against virus attacks.
- The system should have higher speed.
- System should be learnable and usable.ss

3.2 PRELIMINARY INVESTIGATION

The Supermarket Management System is designed to help retail supermarkets streamline and automate their daily activities including inventory management, sales processing, reporting, and customer service. Unlike online shopping systems that focus on e-commerce, this system targets in-store operations and aims to reduce manual work, improve service delivery, and ensure accurate transaction handling within a physical supermarket environment.

While traditional supermarkets struggle with paper-based systems or simple spreadsheets, modern businesses are beginning to adopt digital solutions to eliminate human errors, speed up services, and maintain accurate data across inventory, sales, and employee records. As the competitive market and rising customer expectations grow, supermarkets are learning to improve their service delivery, reduce waste, and enhance customer satisfaction using technology.

A preliminary investigation of the current manual system reveals a number of problems that can be addressed through a computerized supermarket management system. The fact-finding process shows that the supermarket faces the following challenges and opportunities for improvement:

Key Areas of Concern:

1. Improved Customer Service
   – Speeding up billing and checkout processes.
2. Support for Product Tracking
   – Accurate inventory records and low-stock alerts.
3. Better Sales and Performance Reporting
   – Easy generation of daily, weekly, and monthly reports.
4. Information Security Issues
   – Weak controls over manual records and sensitive data.
5. Stronger Administrative Control
   – Better employee monitoring and cash flow tracking.
6. Reduced Operating Costs
   – Less paperwork, fewer errors, and reduced staffing needs.

3.3.1 Organizational Profile

Hayaat Supermarket Management System is a market system which provides various services aimed at streamlining and automating the daily operations of a supermarket. The system offers essential features such as:

Product Management – Adding, editing, and categorizing products in the inventory.

Sales Processing – Recording and managing sales transactions at the cashier level.

Customer Management – Handling customer data and purchases.

Employee Management – Managing staff roles, access levels, and attendance.

Reporting and Analytics – Generating daily, weekly, and monthly sales reports.

Stock Control – Monitoring product stock levels and issuing low-stock alerts. Payment Processing – Supporting multiple payment methods including cash, card, and mobile money.

3.3.2 Current System

Currently, the supermarket relies on a partially manual system for managing sales, inventory, and employee records. This system includes basic POS machines and manual stocktaking methods, which often result in data inconsistencies and delays in service.

3.3.3 Limitations of the Current System

a. Administration Problems
- Lack of real-time data access.
- Errors in manual entry.
- Inefficient report generation.

b. Technical Problems
- Limited system scalability.
- Inadequate integration between sales and inventory.
- Absence of data backup and recovery mechanisms.

3.4 DATA GATHERING

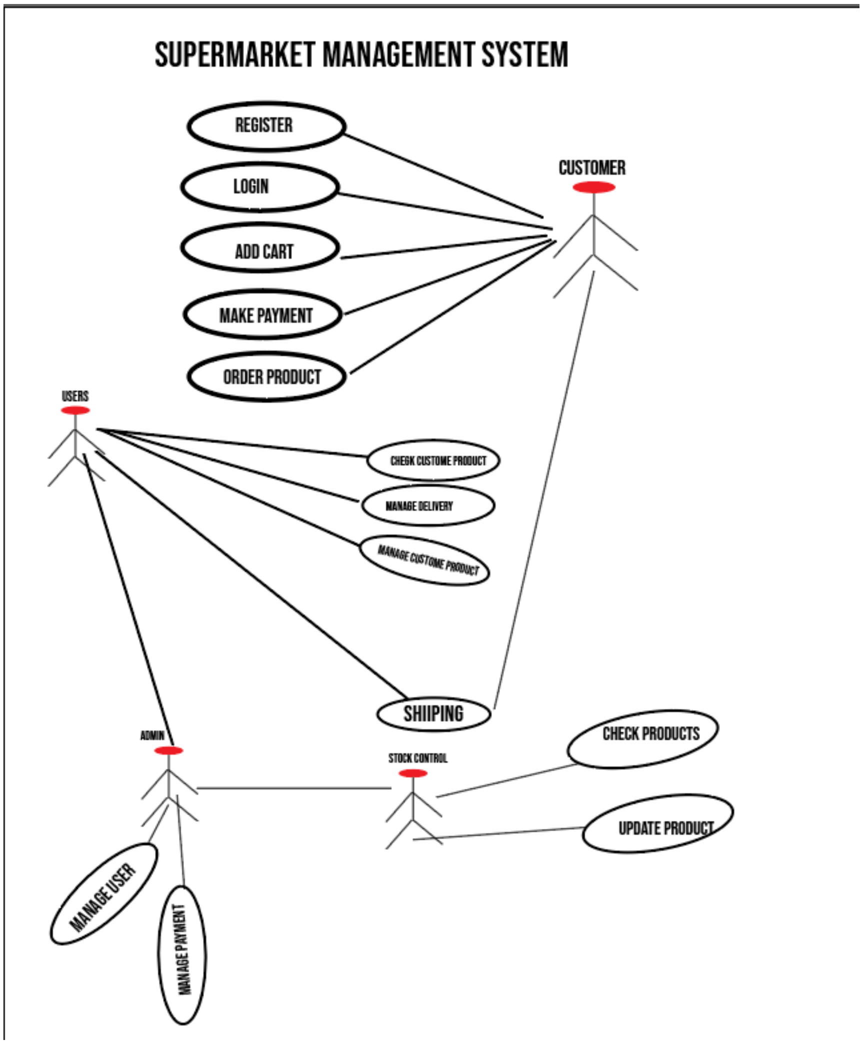
Data gathering was conducted using multiple methodologies to understand the needs and pain points of users. These methods ensured comprehensive coverage of functional and non-functional requirements.

a. Interviews: Conducted with store managers and cashiers to gather insights on daily operations and system expectations.
b. Observations / Surveys: Observed workflows and surveyed customers to understand service delivery gaps.
c. Questionnaires and Other Documents: Collected feedback from stakeholders using structured questionnaires and reviewed existing documentation.

3.4.2 Uniftied Modelling Language

Unified Modeling Language, (UML) is a standardized modeling language consisting of an integrated set of diagrams, developed to help system and software developers for specifying, visualizing, constructing, and documenting the artifacts of software systems, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing object oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects. Using the UML helps project teams communicate, explore potential designs, and validate the architectural design of the software which mostly summarized its features in usability used commonly are use case diagram , activity diagram, class diagram and as well as sequence diagram



3.5 PROBLEM STATEMENT

The current system used by the supermarket is inefficient, prone to errors, and lacks integration between critical business functions. This has resulted in poor customer service, inaccurate data reporting, and challenges in inventory control. A robust, integrated supermarket management system is needed to overcome these challenges and improve overall performance.

3.6 FEASIBILITY STUDY

The feasibility study analyses potential solutions against a set of requirements, evaluates their ability to meet these objectives, describe a recommended solution, and offer a justification for this selection. The feasibility study technical information and cost data to determine the economic potential and practicality of a project. The feasibility study uses techniques that help evaluate a project and/or compare it with other projects. Feasibility study is a part of the system development life cycle, which aims to determine whether it is sensible to develop some system since there are divided into four types below:-

3.6.1 Technical Feasibility

The technically feasibility study basically centers on alternatives for hardware, software and design approach to determine the functional aspects of system relating the system implementing of online Grocery Management System which will be platform independent since it is being coded in PHP (using SUBLIMETEXT and XAMPP).HTML and CSS is used to create online web pages since DBMS should be used for MYSQL database for storing and retrieve data manipulation of scheme system while the hardware and software requirements must be compatible with all operating system (OS) who can operate any computer type of organization

No	Item	Quantity	Unit price t	Total
1	Intel Pentium 4 3.5GHz	3	$270	$810
2	Hp Laser jet 1320 Printer	3	$45	$135
3	Windows 10 Pro	3	$200	$600
Grand Total				**$1,545**

Table 3.6.1. Software & hardware of technical feasible

3.6.2 Operational Feasibility

Operational Feasibility is a measure of how people are able to work with system. This type of feasibility demands if the system will work when developed with their talented skill of professionalism or if organization should demand any alternative solution suggest for recruitment of new staffs or old staffs should be upgraded for training courses regarding their workplace in side of organization as to compact the strategy solution for new proposed system must be forward for implementation of web technology system which has more features including GUI, user friendly interactive and portability with much qualified insurance

No	Human resource cost planning training	Estimated expense costs
1	Training cost for the existing	$40
2	New staff recruitment	$180
3	Training of the new staff	$60
Grand Total		**$280**

Table 3.6.2. Human Resource of Recruitment process

3.6.3 Economic Feasibility Study

Economic analysis is the most frequently used evaluating the effectiveness of proposed system, more commonly known as Benefit analysis or TCO which is to determine benefits and savings that are expected from candidate system and compare them with cost. If the benefits are more than the cost, then decision is made to design and implement the system. The cost and benefits must be tangible or intangible future cost analysis for new proposed system.

No	Assessment TCO Analysis	Expenses Budgets
1	Technical	$1,545
2	Operational	$280
3	Development Cost	$120
4	**Grand Total**	**$1,945**

Table 3.6.3.Economical Assessment Analysis

3.6.3 Schedule Feasibility

Typically this means estimating how long the system will take to develop, and if it can be completed in a given time period using some methods like payback period. Schedule feasibility is a measure of how reasonable the project timetable is. This involves questions such as how much time is available to build the new system, when it can be built , whether it interferes with normal business operation, number of resources required, dependencies, can the system be developed any allocated time for its deadline milestone of project proposal new system

3.7 USER REQUIREMENT SPECIFICATION

In today's digital age, e-commerce applications play a vital role in delivering convenient shopping experiences. The development of a web-based Online Supermarket Management System is essential to provide users—specifically, supermarket customers—with easy and efficient access to grocery shopping and product management.

The system should provide the following core functionalities for supermarket users (customers):

- Create Orders:
- Manage Their Account:
- Login to the System:
- Provide Payment Details:
- Generate Transaction Processes:
- Delivery Service Management

3.7.1 New Proposed System

Our observation case studies were found that mostly was one of the barriers to online shopping adoption has historically been the level of comfort (or should that be discomfort offline system) with technology that causes many consumers' fears of technology were challenged unexpectedly during the pandemic and, since then, we have witnessed increased ease and adoption.

The new proposed online supermarket management system can be platform of bushiness transaction throughout web portal technology which discover and be influenced to purchase products attribute or services based on market place recommendations from friends, peers and trusted sources (like influencers) on social networks like Facebook, Instagram and Twitter which many social media platforms e-online system features, such as in-app checkout, stoppable posts and **" view and Buy Now"** buttons that take users directly to a brand's product page.

The new proposed system is visible and later, will have its usage of online supermarket management system that is able to shop transaction and records of payment method banks or EVC-Plus payment of every transaction that entrance has been it registered and. Also, this system will be implemented using PHP as front end and MySQL(Xampp) as back end

3.7.2 Solution Strategy

After we had discovered the current system's problems, we decided to develop a new system, which meets the needs of the (Hayat Online Supermarket) This solution is associated with implementing sufficient reliable wedding management system, and wiping out the current existing problems; therefore, we considered the solutions and strategies proposed by experts to reach these goals with qualified programmed system.

Table 3.7.1. Solution strategy for criteria programming language

No	Programming Criteria implementation	Selected choice
Option 1	JAVA with ANDROID and PHP with MYSQL	Best option
Option 2	ASP.net with Oracle	Second option
Option 3.	PHP with Mysql	Third option

In our group discussion finally, we decided to **take PHP as front end and MySQL** as back-end in order to develop the new proposed solution strategy for final project graduation in 2024.

3.7.3 System Requirement Specification

Throughout the several meetings conducted with some of the restaurant holders in my

Home town and the problematic statement section, we were able to come up with a set of user stories that must be followed and analyzed during the design phase in order to address for problematic way with possible manner.

In this chapter, we will be listing the different user stories of shoppers with regards to each functional requirement of since the system requirement specification has three dimensions as following below: -

3.7.3.1. System Interface

1. PHP.As Front End
2. MySQL as Back End

3.7.3.2. User interface

1. Login Form
2. Admin Form
3. Shopper's registration Form
4. Manager's form
5. Product item form
6. Shipping form
7. Change Password Form
8. Report Form

3.7.3.3. Software interface

1. Windows XP or Windows 10
2. MySQL with PHP
3. Web browsers

3.4 CHAPTER SUMMARY

This chapter has detailed the requirement analysis process for the supermarket management system. It has covered user needs, system limitations, data gathering methods, feasibility assessment, and the specifications for the proposed system. The findings from this chapter will guide the design and development of a system that meets the operational and strategic goals of the supermarket.

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"The phenomenon is growing fast enough both in prevalence and sophistication that the food industry has coined a name for these combination grocery stores and eateries – the 'grocerant.'"